

## FORWARD DEPLOYED AI ENGINEERING

Capstone Project — CloudServe Solutions

## Product Requirements Document

## Version 2.0

**How to read this document.** This supersedes version 1.0 of 8 September 2026. Requirements marked REVISED changed during the build; requirements marked NEW did not exist in version 1.0. Every change was triggered by a measurement, and the Stage 5 Revision Log records the trigger for each one alongside what version 1.0 said. Nothing here was changed by re-reading the document.

## 1 Document control

| Field                   | Value                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------|
| Version                 | 2.0                                                                                     |
| Date                    | 16 September 2026                                                                       |
| Supersedes              | v1.0 of 8 September 2026                                                                |
| Written by              | Rajarajan Venkatesan                                                                    |
| Status                  | Approved — revised against measured results                                             |
| Evidence base           | Gate run of 14 September 2026, 580 tickets, one unattended run                          |
| Changes in this version | 4 requirements revised (FR-03, FR-04, FR-07, NFR-02), 2 added (FR-13, FR-14), 0 removed |

## 2 Problem statement

CloudServe's support team already has the answers their customers need in their own documentation roughly seven times out of ten, and their agents already know most of the repeat questions by heart. The problem is not a shortage of knowledge. It is that finding and trusting the right answer fast enough, at the moment a ticket first arrives, is broken — so agents fall back on personal habits, escalate things that did not need escalating, and customers wait far longer than the difficulty of their question justifies. Fixing that, rather than replacing the agents' judgement with a chatbot, is what will bring response times and resolution rates back to where the business needs them.

*Unchanged from version 1.0. The build confirmed it: escalation fell from 55.9% to 36.4% and first contact resolution rose from 44.1% to 47.6% using retrieval and routing alone, with no model access at all.*

## 3 Who this is for

| User group                | What they need                                                                                                                                                         | How we will know it worked                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Customers raising tickets | A fast, honest first response — automated where confidence is genuinely high, clearly labelled, with the source cited so they can verify it                            | First-response time drops toward the 2-hour SLA on doc-answerable tickets; no automated response sent to a must_not_auto_respond ticket |
| Tier one support agents   | Trustworthy retrieval against the reviewed corpus, so they stop relying on personal answer files, plus a confidence signal that says when to send and when to escalate | Handling time on 'seen it before' tickets drops from the stated 3–4 minutes of pure search; reliance on personal files falls            |
| Tier two engineers        | Escalations that arrive with a summary, the retrieved sources and a stated reason — not a bare forward                                                                 | Reduced tier-two handling time on tickets arriving with that context                                                                    |
| Head of support           | A measurable rise in first contact resolution, and a system he can explain in a compliance review                                                                      | FCR trending toward the 65% benchmark; every decision carries a logged, readable reason                                                 |

## 4 Functional requirements

Amber rows changed in this version; green rows are new. Each requirement names the discovery evidence behind it and how it is verified.

| ID               | Requirement                                                                                                                                                                                                                                                                                                                                                                              | Priority | Evidence                                                                                              | Verified by                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| FR-01            | The system shall ingest tickets from all four channels (email, chat, docs_comment, forum) and normalise them into one internal representation, preserving original text and channel. Normalisation shall never raise: a malformed ticket yields a normalised object carrying a warning.                                                                                                  | Must     | D-01 channel split; all four are real volume                                                          | A2 — 580/580, zero warnings                                          |
| FR-02            | The system shall classify every ticket's intent and urgency and attach a numeric confidence score, with a defined fallback rather than an exception when it cannot classify confidently.                                                                                                                                                                                                 | Must     | Agent workflow step 2 — judging whether a ticket is 'seen before'                                     | A3 — 80.7% accuracy                                                  |
| FR-03<br>REVISED | The system shall retrieve ranked passages from the 29-article corpus, returning identifiers that resolve to real passages, and shall return nothing rather than a weak match below a relevance threshold of 0.12. The threshold is chosen to maximise recall. ANSWERABILITY IS NOT DECIDED BY THIS SCORE — it is decided by intent, at the routing stage, before the score is consulted. | Must     | D-01 findability, not content, is the failure; threshold sweep 0.00–0.40 showed precision caps at 78% | A4 — hit rate 95.2%                                                  |
| FR-04<br>REVISED | The system shall route each ticket by applying FOUR GATES IN A FIXED ORDER — never-automate intent, safety keyword, no retrieval, then confidence — where the first gate that fires decides. Safety gates are evaluated before the confidence gate. The confidence threshold is 0.45. Routing shall be deterministic and shall record a reason a support manager could read.             | Must     | D-02 47.5% of escalations were doc-answerable; 23 safety leaks under the single-threshold design      | A5 — deterministic; 0 of 101 violations                              |
| FR-05            | The system shall generate answers grounded only in retrieved passages, with citations attached to specific claims, and shall state explicitly when it does not know rather than filling the gap. Without model access the generator is extractive and quotes reviewed articles verbatim.                                                                                                 | Must     | Ines Varga — source attribution; Ravi Menon — 'here is the page it came from'                         | A6 — citation accuracy 92.7%                                         |
| FR-06            | The system shall isolate ticket body text from its own instructions so that customer-supplied text cannot redirect its behaviour.                                                                                                                                                                                                                                                        | Must     | No interview evidence. A necessary consequence of accepting untrusted input                           | Injection test: refund instruction produces neither refund nor claim |
| FR-07<br>REVISED | The system shall run guardrails on every generated response in TWO LAYERS: a deterministic layer that requires no model call and checks five categories (private data, forbidden commitment, unsupported citation, ungrounded answer, false agency), and a model-based semantic layer second. The deterministic layer alone shall be sufficient to block.                                | Must     | The outage test: a guardrail needing the provider is not a guardrail when it is most needed           | A7 — 6 blocks performed with no provider reachable                   |
| FR-08            | The system shall never auto-respond to a ticket whose intent is flagged must_not_auto_respond (security_incident, compliance_request, feature_request, unclear_request) and shall always escalate these regardless of confidence. The list is enforced in code, not in a prompt.                                                                                                         | Must     | Daniel Okonkwo's never-automate list; 101/580 tickets, 100% historically escalated                    | 0 of 101 auto-answered                                               |
| FR-09            | Every automated decision shall be written to a persistent log carrying decision_id, timestamp, ticket_id, stage, prediction, confidence, threshold applied, action taken, plain-language reason, sources used, guardrail results, prompt_version and requirement_ids.                                                                                                                    | Must     | Marcus Adeyemi — autumn compliance review, 'I need to say why it did what it did'                     | A8 — 2,490 records, reconcile exactly                                |
| FR-10            | Escalated tickets shall carry a drafted summary and the retrieved sources considered, so the receiving engineer has context rather than a bare forward.                                                                                                                                                                                                                                  | Must     | Daniel Okonkwo — 'show its working... I would be twice as fast'                                       | Component requirement                                                |
| FR-11            | The system shall provide a single documented command that processes an entire ticket file end to end, unattended, accepting input and output paths as arguments, and shall produce a metrics report automatically.                                                                                                                                                                       | Must     | The hidden grading set is processed through exactly this contract                                     | A1, A9, A10                                                          |

| ID           | Requirement                                                                                                                                                                                                                                                                                                                           | Priority | Evidence                                                                                 | Verified by                                                                                    |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| FR-12        | Every response labelled as automated shall be visibly marked as such to the recipient.                                                                                                                                                                                                                                                | Should   | Ravi Menon — 'that is how I calibrate how much I trust it'                               | Customer-trust requirement                                                                     |
| FR-13<br>NEW | The router shall read the ticket text directly for unambiguous security, legal and billing vocabulary, INDEPENDENTLY of what the classifier concluded, and shall escalate on a match.                                                                                                                                                 | Must     | The classifier alone caught only 72% of must_not_auto_respond tickets                    | 9 tickets escalated by this gate in the gate run — all would otherwise have been auto-answered |
| FR-14<br>NEW | The system shall provide a kill switch that stops automatic answering immediately, without a deployment, checked once per ticket so that it halts a run already in progress. It shall fail closed: if its state cannot be read, automatic answering is disabled. Processing shall continue — the switch stops answering, not the run. | Must     | Governance framework: v1.0's only containment was 'redploy with a changed configuration' | 8 tests in tests/test_kill_switch.py including the fail-closed case                            |

## 5 Non-functional requirements

| ID                | Category     | Requirement                                                                                                                                                                                                                                                                                                  | Verified by                                                                  |
|-------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| NFR-01            | Latency      | Median and 95th-percentile end-to-end latency shall be tracked and reported.                                                                                                                                                                                                                                 | Reported per run — 0.005 s and 0.0061 s on the deterministic path            |
| NFR-02<br>REVISED | Availability | The system shall degrade gracefully rather than crash under provider timeout, outage, rate limiting or malformed input. ADDITIONALLY: after three consecutive provider failures a circuit breaker shall stop outbound calls for the remainder of the run and serve every ticket from the deterministic path. | A11 — induced total outage; 580 tickets completed with 0 errors              |
| NFR-03            | Accuracy     | Classification precision and recall shall be measured per intent class, not only in aggregate.                                                                                                                                                                                                               | Per-class table, 22 classes, in the metrics report                           |
| NFR-04            | Privacy      | No response shall be released containing unredacted private or identifying customer data. Blocking is required; redact-and-send is prohibited.                                                                                                                                                               | 0 private-data detections; customer name is not a field on the ticket object |
| NFR-05            | Auditability | Every decision shall be reconstructable from the decision log alone, without re-running the system.                                                                                                                                                                                                          | A8 — reconciles exactly                                                      |
| NFR-06            | Fairness     | Outcomes shall be measured and reported separately by customer_tier, customer_region and language_fluency.                                                                                                                                                                                                   | Three gaps found and reported: F-01, F-02, F-03                              |
| NFR-07            | Cost         | The system shall run entirely within the free tier of its model provider.                                                                                                                                                                                                                                    | Met trivially — the evaluated path makes no calls at all                     |

## 6 Out of scope

Unchanged from version 1.0, with one addition at the end.

| Not building                                                                           | Why not                                                                                          | What would change it                                               |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Automated responses to feature requests or roadmap questions                           | Zero documentation coverage, confirmed in the data. There is no ground truth to retrieve or cite | A reviewed roadmap knowledge base would have to exist first        |
| Automated responses to genuinely novel incidents                                       | By definition no article exists; this is the category where a wrong answer does most damage      | Not reconsidered within this design — always routed to a human     |
| Automated responses to security, compliance, billing-dispute or data-residency tickets | Daniel Okonkwo's explicit list; 101/580 tickets, 100% historically escalated by humans           | A separately governed workflow with its own sign-off               |
| Consolidating agents' personal answer files                                            | Unreviewed and in places stale. Incorporating them would scale up an existing error              | A formal review bringing them to the standard of the 29 articles   |
| A conversational multi-turn chatbot                                                    | The client's original request. Discovery found it addresses the wrong problem                    | Only as a v2 feature once the retrieval and routing core is proven |
| Rewriting the 29 articles                                                              | They are accurate within the review cycle; the evidenced                                         | If a future audit finds the articles are the                       |

| Not building                                             | Why not                                                                                                                                                                                                                                                | What would change it                                        |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|                                                          | problem is findability                                                                                                                                                                                                                                 | limiting factor                                             |
| <b>A tier-specific routing threshold (ADDED IN v2.0)</b> | Proposed in v1.0 on the assumption that enterprise tickets escalate more. The system reverses that pattern unaided — enterprise now receives the highest automated answer rate of any tier. A tier rule would solve a problem the system does not have | If post-deployment monitoring shows the direction reverting |

## 7 Assumptions, tested

Version 1.0 recorded four assumptions. Two held and two did not. The two that failed changed the architecture.

| Assumption from v1.0                                                           | Held?            | What was found                                                                                                                                                                                                                                                                                  | Consequence                                                                                                                                                                         |
|--------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A confidence threshold set from development data generalises to the hidden set | Partly           | It generalises, but only because the safety gates do not depend on it. One validation ticket still leaked — 'kindly check and revert' read as a deployment rollback, since in Indian English business usage 'revert' means 'reply'. Tuning on visible data did not catch it; held-back data did | Fixed the detector, not the threshold: 'revert' now requires a deployment object. Validation accuracy (83.8%) then exceeded development accuracy (80.2%)                            |
| The 47.5% doc-answerable-but-escalated figure is a gap the system can close    | Yes              | Escalation fell from 55.9% to 36.4% and FCR rose from 44.1% to 47.6%, with no model access at all                                                                                                                                                                                               | Nothing. The assumption was load-bearing and it held — recorded because an assumption that survives contact is worth as much as one that fails                                      |
| Enterprise tickets escalating more (60.4%) is a signal worth designing around  | No               | The historical pattern is real, but the system does not reproduce it. Enterprise receives the HIGHEST automated answer rate of any tier, 69.2% against 61.0% for standard. The direction reversed                                                                                               | Removed the proposed tier-specific threshold. The gap is now reported in the fairness audit and explained rather than engineered around                                             |
| Guardrail blocks are rare enough that always-escalate-on-block is acceptable   | Yes, comfortably | Six blocks across 580 tickets, about 1%, all forbidden commitments. The predicted ceiling was the 17.4% must_not_auto_respond share — an order of magnitude higher                                                                                                                              | Nothing changed. Worth recording that the estimate was wrong in the safe direction, and that the guardrail is not over-triggering, which was the failure mode being guarded against |

## 8 Success measures, against results

| Measure                   | Baseline   | v1.0 target                                  | Measured       | Assessment                                                                                                                                                                                             |
|---------------------------|------------|----------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First contact resolution  | 44.1%      | Toward the 65% benchmark; no number promised | 47.6%          | Moved in the right direction by 3.5 points, well short of 65%. Achieved with no model access, so this is a floor                                                                                       |
| Escalation rate           | 55.9%      | Materially reduced                           | 36.4%          | Met. Down 19.5 points — the largest single movement, and the one that follows directly from the discovery thesis                                                                                       |
| Time to first reply       | 8–12 hours | Materially reduced on doc-answerable tickets | Not comparable | The 0.005 s median is machine time, not queue time. The honest claim is that the lookup step is removed; a deployment would still sit in an operational process                                        |
| Customer satisfaction     | 2.92       | No regression                                | Not measured   | Unmeasurable within scope — no live customers to survey. Flagged as an open question in v1.0 and it remains open. No proxy is offered, because a constructed proxy would look like evidence and not be |
| Never-automate violations | —          | Zero                                         | 0 of 101       | Met, and the measure the project turns on                                                                                                                                                              |

## 9 Open questions

| Question from v1.0                                                                          | Status                    | Resolution or what remains                                                                                                                                                               |
|---------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is CSAT for an automated response measurable within this project's scope?                   | Resolved — no             | It is not. Reported as unmeasured rather than proxied. A shadow deployment with agent review is the only honest route to it                                                              |
| What confidence threshold balances FCR against risk, and must it vary by intent class?      | Resolved                  | 0.45, global. It does not need to vary by class, because class-level exclusion happens at gate one before the threshold is consulted. The question dissolved once the gates were ordered |
| Does chunking strategy materially change retrieval hit rate?                                | Resolved — yes            | Section-aware chunking: 87.7% hit@1 against 85.4% for fixed 800-character windows. More important than the 2.3 points, it never splits a repair sequence                                 |
| Should the enterprise finding change the routing threshold, or only the fairness reporting? | Resolved — reporting only | The system reverses the historical pattern unaided. A tier-specific threshold was rejected and moved to Section 6                                                                        |
| How does the escalation summary get consumed — logged only, or does it need a UI?           | Deferred                  | Logged only within this scope. A deployment would need to answer it; the build specification does not mandate a UI and inventing one was not defensible                                  |
| NEW — Are the three fairness gaps phrasing or structural?                                   | Open                      | F-01, the 10.8 point regional gap, is the largest. Resolving it means segmenting RETRIEVAL SCORES by region, not just outcomes. The data supports the analysis; it has not been done     |
| NEW — Is 69.7% routing accuracy near the ceiling for keyword classification?                | Open                      | Answerable by an error analysis of the 30% that route incorrectly, split into over- and under-escalation. The claim that the shortfall is safe is argued rather than counted             |
| NEW — How does the system perform with model access?                                        | Open                      | The AI path is built, unit-tested and one configuration line away, and has never run against a live provider. Only a working key resolves it                                             |

## 10 Approval

| Role     | Name                 | Date              | Basis                                                   |
|----------|----------------------|-------------------|---------------------------------------------------------|
| Author   | Rajarajan Venkatesan | 16 September 2026 | Revised against the gate run of 14 September            |
| Reviewed | Course coordinator   | 15 September 2026 | Discussed against a working system and measured results |